

TOBYQA: A TIME-AUGMENTED MODEL-BASED METHOD FOR DERIVATIVE-FREE OPTIMIZATION UNDER NOISE AND TEMPORAL DRIFT

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Abstract. Derivative-free optimization (DFO) is challenging when the observation channel varies over time and evaluations are noisy. Conventional model-based methods assume stationary observations; under temporal drift, historical data biases gradient estimates and acceptance tests confound latent-objective decrease with temporal variation. We propose TOBYQA (Time-augmented Optimization BY Quadratic Approximation), a regularized model-based DFO framework that jointly incorporates spatial geometry and temporal variations within a single saddle-point interpolation system. TOBYQA augments the classical least-Frobenius-norm quadratic interpolation system with a linear-in-time drift term and a ridge regularization on the residual kernel, which accommodates noise and ensures well-posedness when the constraint block has full column rank, relaxing the geometric poisedness requirements of classical interpolation. We prove that under affine temporal drift, the recovered gradient is algebraically invariant to the drift rate for any noise scale and sample radius. This property leads to a drift-compensated acceptance test that subtracts the estimated temporal component from the observed reduction. Driven by an adaptive cubic regularization scheme with a closed-form step and a statistical stationarity stopping rule based on the posterior gradient covariance, TOBYQA achieves an oracle complexity of $O(\varepsilon^{-2})$. Benchmark evaluations across the seven evaluation channels show that, at tolerance $\tau = 10^{-3}$, TOBYQA solves 70.2%, 59.9%, and 41.7% of instances at $n = 6, 10$, and 20 , respectively, compared with 46.9%, 27.3%, and 13.8% for the best-performing comparison method. These results demonstrate higher solve rates under temporal drift while retaining comparable performance in static environments.

Key words. derivative-free optimization, time-varying optimization, noisy optimization, cubic regularization, interpolation models, drift compensation

AMS subject classifications. 90C56, 90C30, 65K05

1. Introduction. In some black-box optimization problems, objective evaluations are expensive and depend on when they are performed. Examples include aerodynamic wind-tunnel testing and thermal process control, where the underlying physical configuration $\mathbf{x} \in \mathbb{R}^n$ may have a fixed optimum while sensor drift or temperature fluctuations change the measurement baseline during operation. Because physical trials consume time and resources, evaluation budgets are limited, and earlier evaluations cannot be repeated under identical conditions. A related setting arises in data-driven materials discovery: in the accelerated design of magnetic high-entropy alloys, Li et al. [12] train machine-learning surrogates of alloy properties as functions of the composition vector $\mathbf{x} \in \mathbb{R}^n$ and use the surrogate scores to search the composition space. When such workflows update the surrogate sequentially with newly synthesized batches, the scores supplied to the optimization loop can vary across sequential evaluation rounds.

Motivated by such scenarios, we consider the unconstrained minimization problem

$$(1.1) \quad \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}),$$

where the latent objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous, smooth, and bounded from below by $f_{\text{low}} > -\infty$, but cannot be directly evaluated. Instead, evaluations

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are obtained sequentially through a drifting, noisy observation channel. Queries are indexed by a discrete time $t \in \mathbb{N}$ that increments by one with each oracle evaluation. A query issued at point $\mathbf{x}$ and timestamp t returns

$$(1.2) \quad F(\mathbf{x}, t) = \varphi(f(\mathbf{x}), t) + \varepsilon(\mathbf{x}, t),$$

where $\varphi : \mathbb{R} \times \mathbb{N} \rightarrow \mathbb{R}$ represents an unknown temporal transformation and $\varepsilon(\mathbf{x}, t)$ is an independent zero-mean observation noise with variance bounded by σ_ε^2 . When $\varphi(\cdot, t)$ is strictly increasing with respect to its first argument, the spatial minimizers of f in (1.1) remain invariant across time slices. However, because evaluations are performed sequentially, observations $F(\mathbf{x}_i, t_i)$ collected at distinct time steps reflect disparate temporal baselines. Finite differences and static interpolation models fitted to such data can confound the spatial gradient with temporal variation.

For stationary problems where F depends solely on $\mathbf{x}$, derivative-free optimization (DFO) is well established [8, 1, 11]. Model-based methods such as NEWUOA [19] and BOBYQA [20] construct quadratic surrogates over sparse interpolation sets, commonly with $m = 2n+1$ points, by minimizing the Frobenius norm of the model Hessian change [18]. In noisy settings, Py-BOBYQA [4] provides flexible, noise-aware model management and restart strategies, while the STORM framework [7, 2] establishes global convergence under probabilistically fully-linear models. In dynamic optimization, existing methods mainly target moving optimal trajectories through gradient-based prediction–correction schemes that require temporal derivatives [9, 22], heuristic adaptation in evolutionary algorithms [23], or temporal covariance discounting in Bayesian optimization [14]. These approaches either use derivative information or require more samples than may be available in expensive black-box applications.

Applying classical model-based DFO to time-varying channels raises two difficulties. First, because the interpolation set is accumulated over consecutive iterations, older samples reflect earlier observation baselines; fitting a stationary polynomial surrogate to these space–time observations biases the recovered gradient. Second, the observed difference $F(\mathbf{x}_k, t_k) - F(\mathbf{x}_k + \mathbf{s}_k, t_k + \Delta t)$ combines latent-objective change with the channel variation accumulated over Δt . A step that improves the latent objective may therefore be rejected, while one that worsens it may be accepted. Shrinking the search radius or repeatedly re-evaluating the incumbent consumes additional queries and can reduce the local signal-to-noise ratio.

We propose TOBYQA (Time-augmented Optimization BY Quadratic Approximation), a model-based DFO algorithm that represents spatial geometry and temporal variation in one saddle-point interpolation system. Augmenting the constraint block of the least-Frobenius-norm quadratic system with a time column allows TOBYQA to recover a temporal coefficient γ jointly with the spatial gradient and model constant from a single linear solve. Under affine temporal drift, the recovered gradient is algebraically invariant to the drift rate for any noise scale and sample radius; for general C^2 drift, the gradient bias is bounded by $O(L_\mu \tau_{\max}^2)$. This result motivates a drift-compensated acceptance test that subtracts the estimated temporal increment from the observed reduction. We also show that, under a rotation-invariant curvature prior, Powell’s quadratic interpolation kernel is the covariance kernel of the omitted quadratic terms. This gives a generalized least-squares interpretation of the ridge regularization used to accommodate noise and relax geometric poisedness requirements. The method uses an adaptive cubic regularization scheme with an $O(n)$ closed-form step and a statistical stationarity stopping rule based on the posterior gradient covariance. Under a fully-linear model, TOBYQA achieves oracle complexity $O(\varepsilon^{-2})$. The

benchmark results report higher solve rates under the tested nonstationary channels and performance comparable to classical DFO methods on stationary problems.

2. The TOBYQA Algorithm. This section describes the formulation of the TOBYQA algorithm, detailing the soft-constrained quadratic model, the cubic regularization step, and the algorithmic procedure.

2.1. The Soft-Constrained Quadratic Model. We start with the model-based derivative-free framework used by NEWUOA [19]. At each iteration, NEWUOA maintains m interpolation points around the current iterate and builds a quadratic model $Q \approx f$. The coefficients of Q —the symmetric Hessian, the gradient, and the constant term—have $(n+1)(n+2)/2$ degrees of freedom. Determining all coefficients from the interpolation equations $Q(\mathbf{x}_i) = F_i$, $i = 1, \dots, (n+1)(n+2)/2$, therefore requires a number of evaluations that grows quadratically with n . Powell instead proposed using fewer interpolation points [18]; NEWUOA commonly uses $m = 2n + 1$. The interpolation conditions determine the constant and linear coefficients and provide partial curvature information, while the remaining freedom in the quadratic part is resolved by

$$\min_Q \|\nabla^2 Q - \nabla^2 Q_{\text{prev}}\|_F \quad \text{subject to} \quad Q(\mathbf{x}_i) = F_i, \quad i = 1, \dots, m,$$

which selects, among the quadratics interpolating the samples, the model whose Hessian is closest to that of the previous iteration. Under standard poisedness conditions, this minimum-Frobenius-norm construction yields a unique model. The choice $m = O(n)$ also limits the temporal span of the interpolation set. Theorem 3.7 relates that span to the recovery bias under nonlinear drift.

This interpolation system does not account for the observation channel (1.2). Because the samples are collected at different times, a static quadratic model can absorb temporal variation into its spatial coefficients. In addition, the hard conditions $Q(\mathbf{x}_i) = F_i$ pass observation noise directly into the fitted coefficients. TOBYQA addresses both effects by adding a time column to the model and replacing hard interpolation with a penalized fit.

At iteration k , let $\mathbf{x}_k$ denote the current center and t_k the timestamp at which it was evaluated. The algorithm maintains $m = 2n + 1$ time-stamped samples in the active set $\mathcal{Y}_k$. For each $(\mathbf{x}_i, t_i, F_i) \in \mathcal{Y}_k$, define the spatial displacement $\mathbf{d}_i = \mathbf{x}_i - \mathbf{x}_k$ and temporal offset $\theta_i = t_i - t_k$. The center sample has $\theta_i = 0$; the remaining offsets record the relative query times and may have either sign when the current center is not the most recently evaluated sample. Around the center $\mathbf{x}_k$, we seek a local quadratic-plus-linear-drift surrogate model

$$(2.1) \quad Q_k(\mathbf{x}, t) = c + \mathbf{g}^\top (\mathbf{x} - \mathbf{x}_k) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_k)^\top G (\mathbf{x} - \mathbf{x}_k) + \gamma (t - t_k),$$

with unknowns $c \in \mathbb{R}$, $\mathbf{g} \in \mathbb{R}^n$, symmetric Hessian $G = G^\top \in \mathbb{R}^{n \times n}$, and temporal coefficient $\gamma \in \mathbb{R}$. Here γ describes the local rate of change of the observation channel. Because it is estimated together with $(c, \mathbf{g}, G)$, variation that is linear in time is represented explicitly rather than folded into the spatial model.

The model is determined by the variational problem

$$(2.2) \quad \min_{c, \mathbf{g}, G=G^\top, \gamma} \frac{1}{4} \|G\|_F^2 + \frac{\rho}{2} \sum_{i=1}^m (Q_k(\mathbf{x}_i, t_i) - F_i)^2,$$

where $\rho > 0$ controls the balance between curvature regularization and data fitting. The Frobenius penalty favors a model with modest curvature, while the least-squares

term accommodates noise by allowing the observations to be fitted inexactly. We regularize G itself, rather than its change from the preceding model, because the preceding curvature was fitted on an earlier temporal baseline. With the factor $1/4$, the KKT reduction below produces the classical kernel $A_{ij} = \frac{1}{2}(\mathbf{d}_i^\top \mathbf{d}_j)^2$.

Introduce the residuals $\xi_i := F_i - Q_k(\mathbf{x}_i, t_i)$. Then (2.2) is equivalently written as

$$(2.3) \quad \min_{c, \mathbf{g}, G=G^\top, \gamma, \boldsymbol{\xi}} \quad \frac{1}{4}\|G\|_F^2 + \frac{\rho}{2} \sum_{i=1}^m \xi_i^2 \quad \text{s.t.} \quad Q_k(\mathbf{x}_i, t_i) + \xi_i = F_i, \quad i = 1, \dots, m,$$

Attaching Lagrange multipliers λ_i to the observation constraints yields the Lagrangian

$$(2.4) \quad \mathcal{L}(c, \mathbf{g}, G, \gamma, \boldsymbol{\xi}, \boldsymbol{\lambda}) = \frac{1}{4}\|G\|_F^2 + \frac{\rho}{2} \sum_{i=1}^m \xi_i^2 - \sum_{i=1}^m \lambda_i (Q_k(\mathbf{x}_i, t_i) + \xi_i - F_i).$$

Setting the partial derivatives of $\mathcal{L}$ with respect to the primal variables to zero yields the KKT stationarity conditions:

$$(2.5) \quad \nabla_G \mathcal{L} = 0 \implies G = \sum_{i=1}^m \lambda_i \mathbf{d}_i \mathbf{d}_i^\top,$$

$$(2.6) \quad \nabla_c \mathcal{L} = 0 \implies \sum_{i=1}^m \lambda_i = 0,$$

$$(2.7) \quad \nabla_{\mathbf{g}} \mathcal{L} = 0 \implies \sum_{i=1}^m \lambda_i \mathbf{d}_i = \mathbf{0},$$

$$(2.8) \quad \nabla_{\gamma} \mathcal{L} = 0 \implies \sum_{i=1}^m \lambda_i \theta_i = 0,$$

$$(2.9) \quad \nabla_{\xi_i} \mathcal{L} = 0 \implies \rho \xi_i = \lambda_i \iff \xi_i = \frac{1}{\rho} \lambda_i.$$

The relation $\lambda_i = \rho \xi_i$ ties the multipliers directly to the residuals. The remaining conditions state that $\boldsymbol{\lambda}$ is orthogonal to the constant, spatial-linear, and temporal-linear columns of the model. In particular, $\sum_i \lambda_i \theta_i = 0$ is the extra condition introduced by the time column. It separates a linear temporal component from the spatial fit and leads to the drift-decoupling result in Section 3.

Substituting $\xi_i = \lambda_i/\rho$ and (2.5) into the constraint equations gives, for each $i = 1, \dots, m$, the interpolation relation

$$(2.10) \quad \frac{1}{2} \sum_{j=1}^m \lambda_j (\mathbf{d}_i^\top \mathbf{d}_j)^2 + c + \mathbf{d}_i^\top \mathbf{g} + \gamma \theta_i + \frac{1}{\rho} \lambda_i = F_i.$$

Combining (2.10) with the three orthogonality constraints (2.6), (2.7), and (2.8), we obtain the augmented KKT linear system

$$(2.11) \quad \begin{bmatrix} A + \frac{1}{\rho} I_m & \mathbf{1}_m & D & \boldsymbol{\theta} \\ \mathbf{1}_m^\top & 0 & 0 & 0 \\ D^\top & 0 & 0 & 0 \\ \boldsymbol{\theta}^\top & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ c \\ \mathbf{g} \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ 0 \\ \mathbf{0} \\ 0 \end{bmatrix},$$

whose coefficient matrix we denote by M . Here $A \in \mathbb{R}^{m \times m}$ has entries $A_{ij} = \frac{1}{2}(\mathbf{d}_i^\top \mathbf{d}_j)^2$, $D \in \mathbb{R}^{m \times n}$ has rows $\mathbf{d}_i^\top$, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_m)^\top \in \mathbb{R}^m$, $\mathbf{1}_m$ is the all-ones

vector, and $\mathbf{F} = (F_1, \dots, F_m)^\top$. We denote the augmented constraint block by

$$(2.12) \quad X_{\text{aug}} := [\mathbf{1}_m \quad D \quad \boldsymbol{\theta}] \in \mathbb{R}^{m \times (n+2)},$$

whose first $n + 1$ columns form the constraint block of the classical NEWUOA system, and whose last column $\boldsymbol{\theta}$ provides the temporal extension that enables the joint recovery of the drift rate γ .

2.2. Cubic Regularization Step and Step Acceptance. Given the recovered gradient $\mathbf{g}_k$, TOBYQA computes the trial step from

$$(2.13) \quad \min_{\mathbf{s} \in \mathbb{R}^n} m_k(\mathbf{s}) := \mathbf{g}_k^\top \mathbf{s} + \frac{\sigma_k}{3} \|\mathbf{s}\|_2^3,$$

where $\sigma_k > 0$ controls the step length. Problem (2.13) has the closed-form solution

$$(2.14) \quad \mathbf{s}_k = -\frac{\mathbf{g}_k}{\sqrt{\sigma_k \|\mathbf{g}_k\|_2}}, \quad \|\mathbf{s}_k\|_2 = \sqrt{\frac{\|\mathbf{g}_k\|_2}{\sigma_k}},$$

whose computation requires only $O(n)$ arithmetic operations. The predicted reduction used in the acceptance test below is the decrease of the *linear* part of the model, which excludes the cubic regularization term:

$$(2.15) \quad \text{Pred}_k := -\mathbf{g}_k^\top \mathbf{s}_k = \|\mathbf{g}_k\|_2^{3/2} \sigma_k^{-1/2} = \|\mathbf{g}_k\|_2 \|\mathbf{s}_k\|_2.$$

Evaluating the candidate point $\mathbf{x}_k + \mathbf{s}_k$ through the observation channel at timestamp t_{new} returns $F_{\text{new}} := F(\mathbf{x}_k + \mathbf{s}_k, t_{\text{new}})$. To account for the channel variation over the query interval $\Delta t = t_{\text{new}} - t_k$, we use the recovered temporal coefficient γ to define the *drift-compensated actual reduction*:

$$(2.16) \quad \text{Ared}_k := (F_k - F_{\text{new}}) + \gamma (t_{\text{new}} - t_k),$$

where $F_k := F(\mathbf{x}_k, t_k)$ is the observation at the current center. Since Ared_k contains two independent observation noise terms— $\varepsilon(\mathbf{x}_k, t_k)$ in F_k and $\varepsilon(\mathbf{x}_k + \mathbf{s}_k, t_{\text{new}})$ in F_{new} —its stochastic component has variance $2R_{\text{obs}}$. We therefore use the noise-gated acceptance criterion

$$(2.17) \quad \text{Ared}_k \geq -\frac{\kappa_{\text{tol}}}{2} \sqrt{2R_{\text{obs}}},$$

where $R_{\text{obs}} = \sigma_\varepsilon^2$ is the observation noise variance and $\kappa_{\text{tol}} > 0$ is a stationarity tolerance. When (2.17) holds, the center advances to $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{s}_k$ with $t_{k+1} = t_{\text{new}}$ and $F_{k+1} = F_{\text{new}}$. If the ratio $r_k := \text{Ared}_k / \text{Pred}_k$ exceeds $\eta = 0.75$, the regularization parameter is decreased according to $\sigma_{k+1} = \sigma_k / \gamma_1$, where $\gamma_1 > 1$; otherwise it is held fixed. If (2.17) is violated, the step is rejected, the incumbent center remains unchanged ($(\mathbf{x}_{k+1}, t_{k+1}, F_{k+1}) = (\mathbf{x}_k, t_k, F_k)$), and $\sigma_{k+1} = \gamma_2 \sigma_k$ with $\gamma_2 > 1$.

Each trial evaluation is retained in the active set. To keep its size fixed at $m = 2n + 1$, the sample at the current center is protected and the oldest of the remaining samples is replaced. The resulting temporal span $\tau_{\text{max}} := \max_i |t_i - t_k|$ appears in the nonlinear-drift error bound of Theorem 3.7.

Finally, the algorithm terminates based on a statistical stationarity criterion derived from the posterior covariance of the KKT solve. By Proposition 3.2 below, the primal parameters admit the explicit representation $(c, \mathbf{g}^\top, \gamma)^\top = \Pi \mathbf{F}$ with the recovery operator $\Pi = S^{-1} X_{\text{aug}}^\top K^{-1}$, where $S = X_{\text{aug}}^\top K^{-1} X_{\text{aug}}$ and $K = A + \frac{1}{\rho} I_m$. If the

observation noise were the only error source, i.e., if $\mathbf{F}$ carried the white covariance $R_{\text{obs}}I_m$, the recovered parameters would inherit the covariance

$$(2.18) \quad \text{Cov}(\Pi \mathbf{F}) = R_{\text{obs}} S^{-1} (X_{\text{aug}}^\top K^{-2} X_{\text{aug}}) S^{-1} =: R_{\text{obs}} P_{\text{norm}},$$

a normalized covariance matrix that scales linearly with the noise level; writing $B := X_{\text{aug}}^\top K^{-2} X_{\text{aug}}$, we have $P_{\text{norm}} = S^{-1} B S^{-1}$. In particular, the covariance contribution to the estimated spatial gradient is $R_{\text{obs}}[P_{\text{norm}}]_{gg}$. The algorithm declares convergence when the recovered gradient norm falls below the corresponding uncertainty level:

$$(2.19) \quad \|\mathbf{g}_k\|_2 < \kappa_{\text{tol}} \sqrt{\text{tr}(R_{\text{obs}}[P_{\text{norm}}]_{gg})}.$$

To prevent ill-conditioned sample geometries from inflating the trace of P_{norm} and triggering premature termination, condition (2.19) is evaluated only when $\text{tr}(P_{\text{norm}}) \leq \kappa_{\text{geom}} \cdot \text{tr}(P_{\text{baseline}})$ (with default $\kappa_{\text{geom}} = 5$), where P_{baseline} is computed at the initial coordinate configuration. The iteration also terminates if the trial step stagnates below numerical precision: $\|\mathbf{s}_k\|_2 < 10^{-14} \max(\|\mathbf{x}_k\|_2, 1)$.

2.3. Algorithm Summary. Algorithm 2.1 summarizes the resulting procedure. The default algorithmic parameters are set to $\Delta_0 = 2$, $\gamma_1 = 2$, $\gamma_2 = 1.2$, $\eta = 0.75$, $\kappa_{\text{tol}} = 3$, and $\kappa_{\text{geom}} = 5$. In practical implementation, standard safeguards against floating-point underflow and severe ill-conditioning are applied (e.g., bounding ρ within $[10^2, 10^{14}]$ and lower-bounding scale estimates by machine precision).

Algorithm 2.1. The TOBYQA Algorithm

Require: Initial point $\mathbf{x}_0 \in \mathbb{R}^n$, initial radius $\Delta_0 > 0$, noise variance R_{obs} , parameters $\gamma_1, \gamma_2 > 1$, $\eta \in (0, 1)$, tolerances $\kappa_{\text{tol}}, \kappa_{\text{geom}} > 0$.

Ensure: Approximate stationary point $\mathbf{x}^*$.

- 1: Evaluate initial stencil $\{\mathbf{x}_0\} \cup \{\mathbf{x}_0 \pm \Delta_0 \mathbf{e}_j\}_{j=1}^n$ at sequential timestamps to form $\mathcal{Y}_0$; record initial center $(\mathbf{x}_0, t_0, F_0)$ with $F_0 := F(\mathbf{x}_0, t_0)$.
- 2: Set spread $s_F \leftarrow \max_i F_i - \min_i F_i$, curvature scale $\hat{\sigma}_H \leftarrow s_F / \Delta_0^2$, ridge parameter $\rho \leftarrow \hat{\sigma}_H^2 / R_{\text{obs}}$, and initial cubic weight $\sigma_0 \leftarrow s_F / \Delta_0^3$.
- 3: Compute baseline covariance metric P_{baseline} from the KKT system on $\mathcal{Y}_0$.
- 4: **for** $k = 0, 1, 2, \dots$ **do**
- 5: Construct A , $\boldsymbol{\theta}$, and X_{aug} ; solve (2.11) for $(\boldsymbol{\lambda}, c, \mathbf{g}_k, \gamma_k)$ and compute P_{norm} .
- 6: **if** $\|\mathbf{g}_k\|_2 < \kappa_{\text{tol}} \sqrt{\text{tr}(R_{\text{obs}}[P_{\text{norm}}]_{gg})}$ **and** $\text{tr}(P_{\text{norm}}) \leq \kappa_{\text{geom}} \text{tr}(P_{\text{baseline}})$ **then**
- 7: **return** $\mathbf{x}_k$.
- 8: **end if**
- 9: Compute trial step $\mathbf{s}_k \leftarrow -\mathbf{g}_k / \sqrt{\sigma_k \|\mathbf{g}_k\|_2}$ and predicted reduction $\text{Pred}_k \leftarrow -\mathbf{g}_k^\top \mathbf{s}_k$.
- 10: Query channel at candidate point $(\mathbf{x}_k + \mathbf{s}_k, t_{\text{new}})$ to obtain F_{new} .
- 11: Compute drift-compensated actual reduction $\text{Ared}_k \leftarrow (F_k - F_{\text{new}}) + \gamma_k(t_{\text{new}} - t_k)$ and ratio $r_k \leftarrow \text{Ared}_k / \text{Pred}_k$.
- 12: **if** $\text{Ared}_k \geq -\frac{\kappa_{\text{tol}}}{2} \sqrt{2R_{\text{obs}}}$ **then**
- 13: Set $(\mathbf{x}_{k+1}, t_{k+1}, F_{k+1}) \leftarrow (\mathbf{x}_k + \mathbf{s}_k, t_{\text{new}}, F_{\text{new}})$.
- 14: **if** $r_k > \eta$ **then**
- 15: $\sigma_{k+1} \leftarrow \sigma_k / \gamma_1$.
- 16: **else**
- 17: $\sigma_{k+1} \leftarrow \sigma_k$.
- 18: **end if**
- 19: **else**

20: Set $(\mathbf{x}_{k+1}, t_{k+1}, F_{k+1}) \leftarrow (\mathbf{x}_k, t_k, F_k)$.
 21: $\sigma_{k+1} \leftarrow \gamma_2 \sigma_k$.
 22: **end if**
 23: Remove from $\mathcal{Y}_k$ the oldest sample not located at $\mathbf{x}_{k+1}$.
 24: Insert $(\mathbf{x}_k + \mathbf{s}_k, t_{\text{new}}, F_{\text{new}})$ to form $\mathcal{Y}_{k+1}$.
 25: **end for**

3. Theoretical Properties of the Augmented System. We now analyze the mathematical properties of the time-augmented formulation, including its algebraic solvability, drift decoupling, and statistical interpretation.

3.1. Solvability of the Augmented KKT System. We begin by establishing the algebraic conditions under which the augmented saddle-point KKT system (2.11) possesses a unique solution.

DEFINITION 3.1. *The active interpolation set $\mathcal{Y}_k = \{(\mathbf{x}_i, t_i, F_i)\}_{i=1}^m$ is said to satisfy the augmented column-rank condition if the matrix $X_{\text{aug}} = [\mathbf{1}_m \ D \ \boldsymbol{\theta}] \in \mathbb{R}^{m \times (n+2)}$ has full column rank, i.e.,*

$$(3.1) \quad \text{rank}(X_{\text{aug}}) = n + 2.$$

Condition (3.1) requires that the all-ones vector $\mathbf{1}_m$, the spatial displacement columns $\mathbf{d}_{\cdot,1}, \dots, \mathbf{d}_{\cdot,n}$, and the temporal offset vector $\boldsymbol{\theta}$ are linearly independent in $\mathbb{R}^m$. The following proposition gives the corresponding solvability result.

PROPOSITION 3.2. *Let $K := A + \frac{1}{\rho} I_m \in \mathbb{R}^{m \times m}$, where $A_{ij} = \frac{1}{2}(\mathbf{d}_i^\top \mathbf{d}_j)^2$ and $\rho > 0$. Then:*

- (i) *K is symmetric and strictly positive definite ($K \succ 0$).*
- (ii) *If the active set $\mathcal{Y}_k$ satisfies Definition 3.1, the Schur complement matrix*

$$(3.2) \quad S := X_{\text{aug}}^\top K^{-1} X_{\text{aug}} \in \mathbb{R}^{(n+2) \times (n+2)}$$

is symmetric and strictly positive definite ($S \succ 0$).

- (iii) *The augmented KKT coefficient matrix M in (2.11) is nonsingular, and the unique primal solution is given explicitly by*

$$(3.3) \quad \begin{bmatrix} c \\ \mathbf{g} \\ \gamma \end{bmatrix} = S^{-1} X_{\text{aug}}^\top K^{-1} \mathbf{F} =: \Pi \mathbf{F},$$

where $\Pi := S^{-1} X_{\text{aug}}^\top K^{-1} \in \mathbb{R}^{(n+2) \times m}$ is the linear recovery operator. The corresponding dual variable is $\boldsymbol{\lambda} = K^{-1}(\mathbf{F} - X_{\text{aug}} \Pi \mathbf{F})$.

Proof. For part (i), let $\mathbf{v} \in \mathbb{R}^m$ be non-zero. Since $A_{ij} = \frac{1}{2}(\mathbf{d}_i^\top \mathbf{d}_j)^2$, which equals $\frac{1}{2} \text{tr}((\mathbf{d}_i \mathbf{d}_i^\top)(\mathbf{d}_j \mathbf{d}_j^\top))$, the matrix A is the Gram matrix of the symmetric rank-one matrices $\{\frac{1}{\sqrt{2}} \mathbf{d}_i \mathbf{d}_i^\top\}_{i=1}^m$ under the Frobenius inner product, whence A is symmetric positive semidefinite ($A \succeq 0$). For any $\rho > 0$, we have $\mathbf{v}^\top K \mathbf{v} = \mathbf{v}^\top A \mathbf{v} + \frac{1}{\rho} \|\mathbf{v}\|_2^2 \geq \frac{1}{\rho} \|\mathbf{v}\|_2^2 > 0$, proving $K \succ 0$.

For part (ii), let $\mathbf{u} \in \mathbb{R}^{n+2}$ be non-zero. By Definition 3.1, X_{aug} has full column rank, so $\mathbf{w} := X_{\text{aug}} \mathbf{u} \neq \mathbf{0} \in \mathbb{R}^m$. Because $K \succ 0$, its inverse $K^{-1} \succ 0$. Therefore, $\mathbf{u}^\top S \mathbf{u} = (X_{\text{aug}} \mathbf{u})^\top K^{-1} (X_{\text{aug}} \mathbf{u}) = \mathbf{w}^\top K^{-1} \mathbf{w} > 0$, which establishes $S \succ 0$.

For part (iii), the block 2×2 saddle-point system (2.11) has invertible upper-left block K and invertible Schur complement $-S$. Standard block Gaussian elimination

yields the unique solution: substituting $\boldsymbol{\lambda} = K^{-1}(\mathbf{F} - X_{\text{aug}}[c, \mathbf{g}^\top, \gamma]^\top)$ into $X_{\text{aug}}^\top \boldsymbol{\lambda} = \mathbf{0}$ gives $X_{\text{aug}}^\top K^{-1} X_{\text{aug}}[c, \mathbf{g}^\top, \gamma]^\top = X_{\text{aug}}^\top K^{-1} \mathbf{F}$, which is $S[c, \mathbf{g}^\top, \gamma]^\top = X_{\text{aug}}^\top K^{-1} \mathbf{F}$. Inverting S yields (3.3). $\square$

Remark 3.3. In classical derivative-free optimization (such as NEWUOA [19] or deterministic trust-region methods [8]), interpolation poisedness is used to control both solvability and conditioning. In TOBYQA, the ridge term $\frac{1}{\rho} I_m$ guarantees $K \succ 0$, while Proposition 3.2 reduces the algebraic solvability requirement to full column rank of X_{aug} . The rank condition is weaker than strict Λ -poisedness, although near-rank-deficient configurations may still be poorly conditioned.

COROLLARY 3.4. *Let $\mathcal{Y}_0 = \{(\mathbf{x}_i, t_i, F_i)\}_{i=1}^{2n+1}$ be the initial active set constructed by the coordinate axis-probe with step size $\Delta_0 > 0$ across strictly increasing timestamps $t_1 < t_2 < \dots < t_{2n+1}$, with $\mathbf{x}_1 = \mathbf{x}_0$ evaluated at t_1 . Then $\text{rank}(X_{\text{aug}}) = n + 2$.*

Proof. In the displacement matrix $D \in \mathbb{R}^{(2n+1) \times n}$, the rows corresponding to the probe pair $\mathbf{x}_0 \pm \Delta_0 \mathbf{e}_j$ are $\pm \Delta_0 \mathbf{e}_j^\top$, so the n columns of D are linearly independent and $\text{rank}(D) = n$; moreover, the all-ones column is independent of the columns of D , because the row of D at the center sample is zero while the corresponding entry of $\mathbf{1}_{2n+1}$ is one. Hence $\text{rank}([\mathbf{1}_{2n+1} \ D]) = n + 1$.

Suppose, for contradiction, that $\boldsymbol{\theta} \in \text{span}\{\mathbf{1}_{2n+1}, D\}$, i.e., $\theta_i = \alpha + \beta^\top \mathbf{d}_i$ for some $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R}^n$ and all i . The center row gives $\alpha = \theta_1 = 0$. Applying the same relation to the two rows of the j -th probe pair, whose displacements are $\pm \Delta_0 \mathbf{e}_j$ and whose temporal offsets are θ_{2j} and θ_{2j+1} , gives

$$\theta_{2j} = \alpha + \Delta_0 \beta_j, \quad \theta_{2j+1} = \alpha - \Delta_0 \beta_j \quad \implies \quad \alpha = \frac{1}{2}(\theta_{2j} + \theta_{2j+1}).$$

Since the probe timestamps follow the center timestamp, $\frac{1}{2}(\theta_{2j} + \theta_{2j+1}) > 0$, contradicting $\alpha = 0$. Hence $\boldsymbol{\theta} \notin \text{span}\{\mathbf{1}_{2n+1}, D\}$, and $\text{rank}(X_{\text{aug}}) = n + 2$. $\square$

3.2. Affine-Drift Decoupling in Gradient Recovery. We next examine how an affine temporal component affects the recovered model parameters.

The recovery operator $\Pi = S^{-1} X_{\text{aug}}^\top K^{-1}$ defined in (3.3) satisfies

$$(3.4) \quad \Pi X_{\text{aug}} = (S^{-1} X_{\text{aug}}^\top K^{-1}) X_{\text{aug}} = S^{-1} (X_{\text{aug}}^\top K^{-1} X_{\text{aug}}) = S^{-1} S = I_{n+2}.$$

Thus, Π is a left inverse of the augmented regressor matrix X_{aug} .

THEOREM 3.5. *Suppose the observation channel exhibits affine drift:*

$$(3.5) \quad F(\mathbf{x}, t) = f(\mathbf{x}) + \beta t + \varepsilon(\mathbf{x}, t),$$

for some unknown drift rate $\beta \in \mathbb{R}$. Let $\mathbf{F}^{\text{stat}} := \mathbf{F} - \beta \mathbf{t} \in \mathbb{R}^m$ denote the underlying static observations that would be obtained in the absence of drift, and let $[c^{\text{stat}}, (\mathbf{g}^{\text{stat}})^\top, \gamma^{\text{stat}}]^\top := \Pi \mathbf{F}^{\text{stat}}$ be the model parameters recovered from $\mathbf{F}^{\text{stat}}$. If $\text{rank}(X_{\text{aug}}) = n + 2$, then the parameters recovered by TOBYQA from the drifting vector $\mathbf{F}$ satisfy:

$$(3.6) \quad \mathbf{g} = \mathbf{g}^{\text{stat}}, \quad \gamma = \gamma^{\text{stat}} + \beta, \quad c = c^{\text{stat}} + \beta t_k.$$

Proof. Let $\mathbf{t} = (t_1, \dots, t_m)^\top \in \mathbb{R}^m$ be the vector of query timestamps. Recalling that $t_i = t_k + \theta_i$, we decompose $\mathbf{t}$ over the columns of $X_{\text{aug}} = [\mathbf{1}_m \ D \ \boldsymbol{\theta}]$:

$$(3.7) \quad \mathbf{t} = t_k \mathbf{1}_m + \boldsymbol{\theta} = X_{\text{aug}} \begin{bmatrix} t_k \\ \mathbf{0}_n \\ 1 \end{bmatrix} = X_{\text{aug}} (t_k \mathbf{e}_1 + \mathbf{e}_{n+2}),$$

where $\mathbf{e}_1 = (1, 0, \dots, 0)^\top \in \mathbb{R}^{n+2}$ and $\mathbf{e}_{n+2} = (0, \dots, 0, 1)^\top \in \mathbb{R}^{n+2}$. The observation vector evaluated under (3.5) can be written as $\mathbf{F} = \mathbf{F}^{\text{stat}} + \beta \mathbf{t}$. Applying the linear recovery operator Π to $\mathbf{F}$ and invoking the left-inverse identity (3.4), we obtain:

$$\begin{aligned} \begin{bmatrix} c \\ \mathbf{g} \\ \gamma \end{bmatrix} &= \Pi \mathbf{F} = \Pi \mathbf{F}^{\text{stat}} + \beta \Pi \mathbf{t} \\ &= \begin{bmatrix} c^{\text{stat}} \\ \mathbf{g}^{\text{stat}} \\ \gamma^{\text{stat}} \end{bmatrix} + \beta \Pi X_{\text{aug}} (t_k \mathbf{e}_1 + \mathbf{e}_{n+2}) \\ &= \begin{bmatrix} c^{\text{stat}} \\ \mathbf{g}^{\text{stat}} \\ \gamma^{\text{stat}} \end{bmatrix} + \beta (t_k \mathbf{e}_1 + \mathbf{e}_{n+2}) \\ &= \begin{bmatrix} c^{\text{stat}} + \beta t_k \\ \mathbf{g}^{\text{stat}} \\ \gamma^{\text{stat}} + \beta \end{bmatrix}. \end{aligned}$$

Equating respective components yields (3.6). $\square$

Remark 3.6. Theorem 3.5 states that, for any drift rate β , any noise realization, and any sample geometry satisfying the augmented rank condition, adding the affine term $\beta \mathbf{t}$ does not change the recovered spatial gradient. The result does not depend on the particular Powell kernel used here; it requires only that K be symmetric positive definite, so that $\Pi X_{\text{aug}} = I_{n+2}$. In regression terms, this is an instance of the Frisch–Waugh–Lovell principle [10, 13]: the temporal column is treated as a nuisance regressor, and the spatial coefficients are recovered after accounting for that column.

We next generalize the analysis to nonlinear observation drift $\mu(t)$ with bounded second derivative.

THEOREM 3.7. *Suppose the observation channel has the form $F(\mathbf{x}, t) = f(\mathbf{x}) + \mu(t) + \varepsilon(\mathbf{x}, t)$, where $\mu \in C^2(\mathbb{R})$ with $|\mu''(t)| \leq L_\mu$ for all t . Let $\tau_{\max} := \max_i |t_i - t_k|$ denote the maximum temporal span of the active set $\mathcal{Y}_k$. Then:*

$$(3.8) \quad |\gamma - \gamma^{\text{stat}} - \mu'(t_k)| \leq C_\gamma L_\mu \tau_{\max}, \quad \|\mathbf{g} - \mathbf{g}^{\text{stat}}\|_2 \leq C_g L_\mu \tau_{\max}^2,$$

where, for a fixed normalized sample geometry, the constants $C_\gamma, C_g > 0$ are independent of $\tau_{\max}$ and L_μ ; they depend on the normalized spatial and temporal configuration and on ρ .

Proof. Performing a second-order Taylor expansion of $\mu(t_i)$ around the center timestamp t_k gives

$$\mu(t_i) = \mu(t_k) + \mu'(t_k) \theta_i + r_\mu(t_i), \quad |r_\mu(t_i)| \leq \frac{1}{2} L_\mu \theta_i^2 \leq \frac{1}{2} L_\mu \tau_{\max}^2.$$

In vector notation,

$$\boldsymbol{\mu} = \mu(t_k) \mathbf{1}_m + \mu'(t_k) \boldsymbol{\theta} + \mathbf{r}_\mu = X_{\text{aug}} [\mu(t_k) \mathbf{e}_1 + \mu'(t_k) \mathbf{e}_{n+2}] + \mathbf{r}_\mu.$$

Applying the recovery operator Π , the constant and linear terms in time are represented by the corresponding columns of X_{aug} , leaving the perturbation induced by the second-order residual $\mathbf{r}_\mu$:

$$\begin{bmatrix} c - c^{\text{stat}} - \mu(t_k) \\ \mathbf{g} - \mathbf{g}^{\text{stat}} \\ \gamma - \gamma^{\text{stat}} - \mu'(t_k) \end{bmatrix} = \Pi \mathbf{r}_\mu.$$

It remains to bound the components of $\Pi \mathbf{r}_\mu$. Write $\tau := \tau_{\max}$ and factor the temporal column as $\boldsymbol{\theta} = \tau \hat{\boldsymbol{\theta}}$ with $\|\hat{\boldsymbol{\theta}}\|_\infty \leq 1$. With

$$T := [\mathbf{1}_m \quad D \quad \hat{\boldsymbol{\theta}}], \quad J := \text{diag}(I_{n+1}, \tau), \quad \text{so that} \quad X_{\text{aug}} = TJ,$$

we have $S = X_{\text{aug}}^\top K^{-1} X_{\text{aug}} = J \hat{S} J$ with $\hat{S} := T^\top K^{-1} T$. The matrix $K = A + \rho^{-1} I_m$ depends only on the spatial displacements and ρ , so both K and $\hat{S}$ are independent of τ under this scaling, and

$$\Pi \mathbf{r}_\mu = S^{-1} X_{\text{aug}}^\top K^{-1} \mathbf{r}_\mu = J^{-1} \hat{S}^{-1} T^\top K^{-1} \mathbf{r}_\mu,$$

where $J^{-1} = \text{diag}(I_{n+1}, \tau^{-1})$. Since the matrix $\hat{S}^{-1} T^\top K^{-1}$ is independent of τ , the scaling of the components of $\Pi \mathbf{r}_\mu = J^{-1} (\hat{S}^{-1} T^\top K^{-1} \mathbf{r}_\mu)$ is governed by J^{-1} : the $(c, \mathbf{g})$ -components carry factor 1, while the γ -component carries factor τ^{-1} . Combining this with $\|\mathbf{r}_\mu\|_2 \leq \sqrt{m} \|\mathbf{r}_\mu\|_\infty \leq \frac{\sqrt{m}}{2} L_\mu \tau^2$ yields

$$\begin{aligned} |\gamma - \gamma^{\text{stat}} - \mu'(t_k)| &\leq \frac{\|\hat{S}^{-1} T^\top K^{-1}\|_2 \sqrt{m}}{2} L_\mu \tau, \\ \|\mathbf{g} - \mathbf{g}^{\text{stat}}\|_2 &\leq \frac{\|\hat{S}^{-1} T^\top K^{-1}\|_2 \sqrt{m}}{2} L_\mu \tau^2, \end{aligned}$$

which is (3.8) with $C_\gamma = C_g = \frac{\sqrt{m}}{2} \|\hat{S}^{-1} T^\top K^{-1}\|_2$, a quantity independent of τ and L_μ . $\square$

Remark 3.8. Theorem 3.7 shows that the temporal span of the active set directly controls the nonlinear-drift contribution to the recovered gradient. The time-priority replacement rule in Section 2.2 discards the oldest non-center sample and therefore keeps the retained observations temporally compact, subject to the protected center point.

3.3. Statistical Interpretation of the Powell Residual Kernel. We next derive a statistical interpretation of the kernel matrix $K = A + \frac{1}{\rho} I_m$.

Recall from Section 2.2 that the step computation uses the model with $H_k = 0$, so the surrogate minimized in the trial step subproblem is linear in space: $Q_k(\mathbf{d}, \theta) = c + \mathbf{g}^\top \mathbf{d} + \gamma \theta$. A second-order Taylor expansion of f about $\mathbf{x}_k$ then gives, for each sample,

$$(3.9) \quad F_i - Q_k(\mathbf{d}_i, \theta_i) = \frac{1}{2} \mathbf{d}_i^\top H \mathbf{d}_i + O(\|\mathbf{d}_i\|_2^3) + \varepsilon_i,$$

where $H = \nabla^2 f(\mathbf{x}_k)$ is the local Hessian omitted from the decision model and $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is white observation noise. After the zeroth- and first-order terms and the temporal component have been represented by $(c, \mathbf{g}, \gamma)$, spatial curvature enters the residual quadratically in the displacement. The following proposition gives the covariance of these quadratic terms under a rotation-invariant prior and relates it to Powell's kernel A .

PROPOSITION 3.9. *Let $H \in \mathbb{R}^{n \times n}$ be a random symmetric matrix drawn from a rotation-invariant ensemble with $\mathbb{E}[H] = 0$ and*

$$(3.10) \quad \mathbb{E}[H_{ab} H_{cd}] = \sigma_H^2 (\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc}),$$

where $\sigma_H^2 > 0$ reflects the curvature uncertainty scale. Then for any displacement vectors $\mathbf{d}_i, \mathbf{d}_j \in \mathbb{R}^n$, the covariance between the unmodeled quadratic forms $q_i := \frac{1}{2} \mathbf{d}_i^\top H \mathbf{d}_i$ and $q_j := \frac{1}{2} \mathbf{d}_j^\top H \mathbf{d}_j$ is given by

$$(3.11) \quad \text{Cov}(q_i, q_j) = \frac{\sigma_H^2}{2} (\mathbf{d}_i^\top \mathbf{d}_j)^2 = \sigma_H^2 A_{ij}.$$

Proof. Writing $q_i = \frac{1}{2} \sum_{a,b=1}^n H_{ab} d_{i,a} d_{i,b}$, we have $\mathbb{E}[q_i] = 0$. The covariance is:

$$\begin{aligned} \text{Cov}(q_i, q_j) &= \mathbb{E}[q_i q_j] = \frac{1}{4} \sum_{a,b=1}^n \sum_{c,d=1}^n \mathbb{E}[H_{ab} H_{cd}] d_{i,a} d_{i,b} d_{j,c} d_{j,d} \\ &= \frac{\sigma_H^2}{4} \sum_{a,b,c,d=1}^n (\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc}) d_{i,a} d_{i,b} d_{j,c} d_{j,d} \\ &= \frac{\sigma_H^2}{4} \left(\sum_{a,b=1}^n d_{i,a} d_{j,a} d_{i,b} d_{j,b} + \sum_{a,b=1}^n d_{i,a} d_{j,b} d_{i,b} d_{j,a} \right) \\ &= \frac{\sigma_H^2}{4} \left((\mathbf{d}_i^\top \mathbf{d}_j)^2 + (\mathbf{d}_i^\top \mathbf{d}_j)^2 \right) = \frac{\sigma_H^2}{2} (\mathbf{d}_i^\top \mathbf{d}_j)^2 = \sigma_H^2 A_{ij}, \end{aligned}$$

where the last step uses $A_{ij} = \frac{1}{2} (\mathbf{d}_i^\top \mathbf{d}_j)^2$. Hence the covariance of the omitted quadratic terms is $\sigma_H^2 A$. $\square$

Under Proposition 3.9, treating the unmodeled curvature as a Gaussian random effect with covariance $\sigma_H^2 A$ and observation noise as independent Gaussian with covariance $\sigma_\varepsilon^2 I_m$, the total covariance of the residual vector is

$$\Sigma_{\text{res}} = \sigma_H^2 A + \sigma_\varepsilon^2 I_m = \sigma_H^2 \left(A + \frac{\sigma_\varepsilon^2}{\sigma_H^2} I_m \right) = \sigma_H^2 (A + \rho^{-1} I_m) = \sigma_H^2 K,$$

where $\rho = \sigma_H^2 / \sigma_\varepsilon^2$ is the signal-to-noise ratio between curvature variation and observation noise. The generalized least squares (GLS) estimator of $[c, \mathbf{g}^\top, \gamma]^\top$ under residual covariance Σ_{res} is $(X_{\text{aug}}^\top \Sigma_{\text{res}}^{-1} X_{\text{aug}})^{-1} X_{\text{aug}}^\top \Sigma_{\text{res}}^{-1} \mathbf{F} = (X_{\text{aug}}^\top K^{-1} X_{\text{aug}})^{-1} X_{\text{aug}}^\top K^{-1} \mathbf{F} = \Pi \mathbf{F}$.

Thus, under this residual model, the soft-constrained KKT estimate (2.11) coincides with the generalized least-squares estimator.

3.4. Drift-Compensated Reduction and Error Decomposition. Finally, we analyze the drift-compensated actual reduction Ared_k defined in (2.16).

PROPOSITION 3.10. *Suppose queries are performed through the channel $F(\mathbf{x}, t) = \phi(f(\mathbf{x}), t) + \varepsilon(\mathbf{x}, t)$. For a trial step $\mathbf{s}_k$ evaluated at t_{new} with center $\mathbf{x}_k$ at t_k , the actual reduction has the decomposition*

$$(3.12) \quad \text{Ared}_k = \Delta f(\mathbf{x}_k, \mathbf{s}_k) + \mathcal{E}_{\text{drift}} + \mathcal{E}_{\text{noise}},$$

where:

- (i) $\Delta f(\mathbf{x}_k, \mathbf{s}_k) := f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{s}_k)$ is the latent-objective reduction.
- (ii) $\mathcal{E}_{\text{drift}} := \gamma_k(t_{\text{new}} - t_k) - (\phi(f(\mathbf{x}_k + \mathbf{s}_k), t_{\text{new}}) - \phi(f(\mathbf{x}_k), t_k)) - \Delta f(\mathbf{x}_k, \mathbf{s}_k)$ is the temporal compensation error.
- (iii) $\mathcal{E}_{\text{noise}} := \varepsilon(\mathbf{x}_k, t_k) - \varepsilon(\mathbf{x}_k + \mathbf{s}_k, t_{\text{new}})$ is the zero-mean observation noise error with variance $2\sigma_\varepsilon^2$.

In particular, under affine drift $\phi(f(\mathbf{x}), t) = f(\mathbf{x}) + \beta t$, Theorem 3.5 gives

$$(3.13) \quad \mathcal{E}_{\text{drift}} = (\gamma_k - \beta)(t_{\text{new}} - t_k) = \gamma_k^{\text{stat}}(t_{\text{new}} - t_k).$$

Thus the contribution of the affine drift rate β cancels from $\mathcal{E}_{\text{drift}}$. A residual offset may remain because the drift-free spatial data can induce a nonzero coefficient γ_k^{stat} through space-time correlation in the sequential sample geometry.

Proof. Substituting the channel expressions for F_k and F_{new} into (2.16) and grouping the spatial, temporal, and noise terms yields (3.12). Under affine drift, the temporal difference contributes $\beta(t_{\text{new}} - t_k)$, while Theorem 3.5 gives $\gamma_k = \gamma_k^{\text{stat}} + \beta$. Substitution yields (3.13). $\square$

4. Convergence and Complexity Analysis. In this section, we establish the convergence properties and oracle complexity of TOBYQA under fully-linear models.

4.1. Standing Assumptions and Stochastic Framework. We analyze TOBYQA under the classical framework of fully-linear models [8, 7, 4]. Let $\mathcal{F}_{k-1} := \sigma(\mathbf{x}_0, \dots, \mathbf{x}_k, \mathcal{Y}_0, \dots, \mathcal{Y}_k)$ denote the filtration representing the history of the algorithm up to the beginning of iteration k . We impose the following standard assumptions on the latent objective function and the observation channel.

ASSUMPTION 4.1. *The objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable with L_g -Lipschitz continuous gradient:*

$$(4.1) \quad \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L_g \|\mathbf{x} - \mathbf{y}\|_2, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n,$$

and is bounded from below by $f_{\text{low}} > -\infty$ on $\mathbb{R}^n$.

ASSUMPTION 4.2. (Active-set radius bounds.) *There exist constants $0 < \delta_{\text{lo}} \leq \delta_{\text{hi}} < \infty$ such that, at every iteration prior to termination,*

$$(4.2) \quad \delta_{\text{lo}} \leq \delta_k \leq \delta_{\text{hi}}, \quad \delta_k := \max_{i=1, \dots, m} \|\mathbf{x}_i - \mathbf{x}_k\|_2.$$

Such active-set radius bounds are standard in derivative-free complexity analysis [8, 4].

ASSUMPTION 4.3. (Fully-linear gradient accuracy.) *There exists a constant $\kappa_{eg} > 0$ such that, at every iteration $k \geq 0$, the recovered model gradient $\mathbf{g}_k$ satisfies*

$$(4.3) \quad \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \leq \kappa_{eg} \zeta_k,$$

where $\zeta_k := \max\{\delta_k, \sigma_\varepsilon/\delta_k\}$ and δ_k is the active sample radius in (4.2).

Assumption 4.3 is the standard accuracy condition for fully-linear models in derivative-free optimization [8]. Its error scale combines the sample radius δ_k , which controls spatial truncation, and the ratio $\sigma_\varepsilon/\delta_k$, which accounts for noise amplification.

For the observation channel, Proposition 3.10 and Theorem 3.5 show that the affine drift rate is represented by the temporal coefficient and does not enter the recovered spatial gradient. They do not imply that the drift-free time coefficient γ_k^{stat} vanishes. Sequential space-time correlation can produce a nonzero γ_k^{stat} even without observation noise. We therefore define the conditional acceptance bias

$$(4.4) \quad b_k := \mathbb{E}[\text{Ared}_k \mid \mathcal{F}_{k-1}] - (f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{s}_k)).$$

Under affine drift, relation (3.13) identifies the deterministic space-time aliasing contribution as $\gamma_k^{\text{stat}} \Delta t_k$; under C^2 drift, b_k also contains the nonlinear temporal truncation error. The following assumption controls this conditional bias and the remaining stochastic innovation.

ASSUMPTION 4.4. (Acceptance-channel residual.) *There exists $\varepsilon > 0$ such that, whenever the incumbent satisfies $\|\nabla f(\mathbf{x}_k)\|_2 \geq \varepsilon$, the conditional bias in (4.4) obeys*

$$(4.5) \quad |b_k| \leq \frac{1}{4} \text{Pred}_k,$$

and the centered innovation $\text{Ared}_k - \mathbb{E}[\text{Ared}_k \mid \mathcal{F}_{k-1}]$ is conditionally sub-Gaussian with variance proxy $2\sigma_\varepsilon^2$.

4.2. Reliable Regularization Regime and Step Decrease. Recall from Section 2.2 that the Zero-Hessian closed-form cubic trial step $\mathbf{s}_k$ and the corresponding predicted reduction Pred_k satisfy:

$$(4.6) \quad \mathbf{s}_k = -\frac{\mathbf{g}_k}{\sqrt{\sigma_k \|\mathbf{g}_k\|_2}}, \quad \text{Pred}_k = \|\mathbf{g}_k\|_2^{3/2} \sigma_k^{-1/2} = \|\mathbf{g}_k\|_2 \|\mathbf{s}_k\|_2,$$

with $\|\mathbf{s}_k\|_2 = \sqrt{\|\mathbf{g}_k\|_2 / \sigma_k}$. We first give a sufficient lower bound on the cubic regularization weight σ_k for decrease in the latent objective f .

LEMMA 4.5. (Descent under sufficient regularization.) *Let Assumption 4.1 hold. If the cubic regularization weight satisfies*

$$(4.7) \quad \sigma_k \geq \sigma^*(\mathbf{g}_k) := \frac{4L_g^2}{\|\mathbf{g}_k\|_2},$$

then the objective decrease along the candidate trial step $\mathbf{s}_k$ satisfies

$$(4.8) \quad f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{s}_k) \geq \frac{3}{4} \text{Pred}_k - \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \|\mathbf{s}_k\|_2.$$

Proof. By the classical descent lemma for L_g -smooth functions (Assumption 4.1), we have

$$\begin{aligned} f(\mathbf{x}_k + \mathbf{s}_k) &\leq f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^\top \mathbf{s}_k + \frac{L_g}{2} \|\mathbf{s}_k\|_2^2 \\ &= f(\mathbf{x}_k) + \mathbf{g}_k^\top \mathbf{s}_k + \frac{L_g}{2} \|\mathbf{s}_k\|_2^2 + (\nabla f(\mathbf{x}_k) - \mathbf{g}_k)^\top \mathbf{s}_k. \end{aligned}$$

Using $\mathbf{g}_k^\top \mathbf{s}_k = -\|\mathbf{g}_k\|_2 \|\mathbf{s}_k\|_2 = -\text{Pred}_k$ and Cauchy-Schwarz, rearranging terms yields

$$(4.9) \quad f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{s}_k) \geq \text{Pred}_k - \frac{L_g}{2} \|\mathbf{s}_k\|_2^2 - \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \|\mathbf{s}_k\|_2.$$

We now evaluate the ratio of the quadratic curvature penalty $\frac{L_g}{2} \|\mathbf{s}_k\|_2^2$ to the predicted decrease Pred_k . Substituting the closed-form identities (4.6) gives:

$$(4.10) \quad \frac{\frac{L_g}{2} \|\mathbf{s}_k\|_2^2}{\text{Pred}_k} = \frac{\frac{L_g}{2} (\|\mathbf{g}_k\|_2 / \sigma_k)}{\|\mathbf{g}_k\|_2^{3/2} \sigma_k^{-1/2}} = \frac{L_g}{2\sqrt{\sigma_k \|\mathbf{g}_k\|_2}}.$$

Under condition (4.7), we have $\sigma_k \|\mathbf{g}_k\|_2 \geq 4L_g^2$, which implies $\sqrt{\sigma_k \|\mathbf{g}_k\|_2} \geq 2L_g$. Substituting this inequality into (4.10) yields

$$\frac{\frac{L_g}{2} \|\mathbf{s}_k\|_2^2}{\text{Pred}_k} \leq \frac{L_g}{2(2L_g)} = \frac{1}{4}, \quad \text{whence} \quad \frac{L_g}{2} \|\mathbf{s}_k\|_2^2 \leq \frac{1}{4} \text{Pred}_k.$$

Substituting $\frac{L_g}{2} \|\mathbf{s}_k\|_2^2 \leq \frac{1}{4} \text{Pred}_k$ into (4.9) yields (4.8), completing the proof. $\square$

Remark 4.6. (Interpretation of the regularization scale.) Condition (4.7) identifies the scale at which the curvature term in the descent bound is at most one quarter of the predicted reduction. In Algorithm 2.1, changing σ_k changes the step length through $\|\mathbf{s}_k\|_2 = \sqrt{\|\mathbf{g}_k\|_2 / \sigma_k}$: rejection increases σ_k , whereas an accepted step with $r_k > \eta$ decreases it. Thus σ_k controls the step length without a separately maintained trust-region radius.

4.3. Rejection Probability and Regularization Confinement.

LEMMA 4.7. **(Rejection probability in the reliable regime.)** *Let Assumption 4.1 hold, let the residual drift bound (4.5) hold at iteration k , and suppose that*

$$(4.11) \quad \sigma_k \geq \gamma_2 \sigma^*(\mathbf{g}_k) \quad \text{and} \quad \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \|\mathbf{s}_k\|_2 \leq \frac{1}{4} \text{Pred}_k.$$

Then the conditional rejection probability obeys

$$(4.12) \quad \mathbb{P}(\text{iteration } k \text{ is rejected} \mid \mathcal{F}_{k-1}) \leq e^{-\kappa_{\text{tol}}^2/8} =: p',$$

and $p' \leq \frac{1}{2}$ whenever $\kappa_{\text{tol}} \geq \sqrt{8 \ln 2}$.

Proof. Under $\sigma_k \geq \gamma_2 \sigma^*(\mathbf{g}_k)$ we have $\sigma_k \|\mathbf{g}_k\|_2 \geq 4L_g^2$, so Lemma 4.5 gives $\Delta f_k \geq \frac{3}{4} \text{Pred}_k - \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \|\mathbf{s}_k\|_2 \geq \frac{1}{2} \text{Pred}_k$. Together with (4.5) and the expectation (4.4), this yields

$$\mathbb{E}[\text{Ared}_k \mid \mathcal{F}_{k-1}] = \Delta f_k + b_k \geq \frac{1}{2} \text{Pred}_k - \frac{1}{4} \text{Pred}_k = \frac{1}{4} \text{Pred}_k > 0.$$

The deviation $\text{Ared}_k - \mathbb{E}[\text{Ared}_k \mid \mathcal{F}_{k-1}]$ is sub-Gaussian with variance proxy $2\sigma_\varepsilon^2 = 2R_{\text{obs}}$. Hence, by the sub-Gaussian tail bound and the noise gate (2.17),

$$\begin{aligned} \mathbb{P}(\text{Ared}_k < -\frac{\kappa_{\text{tol}}}{2} \sqrt{2R_{\text{obs}}} \mid \mathcal{F}_{k-1}) &\leq \exp\left(-\frac{(\frac{\kappa_{\text{tol}}}{2} \sqrt{2R_{\text{obs}}} + \frac{1}{4} \text{Pred}_k)^2}{4R_{\text{obs}}}\right) \\ &\leq \exp\left(-\frac{\frac{\kappa_{\text{tol}}^2}{4} 2R_{\text{obs}}}{4R_{\text{obs}}}\right) = e^{-\kappa_{\text{tol}}^2/8}, \end{aligned}$$

where the last inequality uses $\text{Pred}_k > 0$, and $p' \leq \frac{1}{2}$ for $\kappa_{\text{tol}} \geq \sqrt{8 \ln 2}$ since $e^{-\ln 2} = \frac{1}{2}$. $\square$

Lemma 4.7 demonstrates that in the reliable regularization regime, trial step rejections are rare events with exponentially decaying probability $p' \leq e^{-\kappa_{\text{tol}}^2/8} \leq \frac{1}{2}$. While this guarantees frequent successful steps in practice, establishing an almost-sure upper bound on σ_k solely from this mechanism would require uniform geometric replenishment and lower bounds on $\mathbb{P}(r_k > \eta)$, which are difficult to verify a priori under arbitrary temporal drift. We therefore adopt regularization confinement as a standing assumption.

ASSUMPTION 4.8. **(Regularization confinement.)** *For the target accuracy $\varepsilon > 0$ with $\varepsilon \leq 8L_g^2/\sigma_0$ and $\kappa_{\text{tol}} \geq \sqrt{8 \ln 2}$, the regularization parameter satisfies*

$$(4.13) \quad \sigma^*(\mathbf{g}_k) \leq \sigma_k \leq \sigma_{\max} := \gamma_2 \frac{8L_g^2}{\varepsilon},$$

throughout all iterations k prior to termination, i.e., while $\min_{j \leq k} \|\nabla f(\mathbf{x}_j)\|_2 \geq \varepsilon$.

Complementing the probabilistic rejection bound of Lemma 4.7, the following proposition provides a deterministic upper bound on the number of unsuccessful iterations along any sample trajectory under regularization confinement.

PROPOSITION 4.9. **(Bound on unsuccessful iterations.)** *Let $\mathcal{S}_K := \{k \leq K : \text{iteration } k \text{ is accepted}\}$ and $\mathcal{U}_K := \{k \leq K : \text{iteration } k \text{ is rejected}\}$ denote the index sets of successful and unsuccessful iterations up to iteration K , respectively. Then:*

$$(4.14) \quad |\mathcal{U}_K| \leq \frac{\log \gamma_1}{\log \gamma_2} |\mathcal{S}_K| + \frac{1}{\log \gamma_2} \log \frac{\sigma_{\max}}{\sigma_0}.$$

Proof. From the update rule in Algorithm 2.1, the weight decreases by γ_1 only on accepted iterations with $r_k > \eta$, is held fixed on accepted iterations with $r_k \leq \eta$, and increases by γ_2 on rejected iterations. Let $N_1 \leq |\mathcal{S}_K|$ denote the number of accepted iterations on which $r_k > \eta$. Taking logarithms and telescoping from $k = 0$ to $K - 1$ gives

$$\log \sigma_K = \log \sigma_0 - N_1 \log \gamma_1 + |\mathcal{U}_K| \log \gamma_2.$$

Since $N_1 \leq |\mathcal{S}_K|$ and $\sigma_K \leq \sigma_{\max}$ by Assumption 4.8, rearranging yields

$$|\mathcal{U}_K| \log \gamma_2 = \log \frac{\sigma_K}{\sigma_0} + N_1 \log \gamma_1 \leq \log \frac{\sigma_{\max}}{\sigma_0} + |\mathcal{S}_K| \log \gamma_1,$$

and dividing by $\log \gamma_2 > 0$ establishes (4.14). $\square$

4.4. Iteration and Oracle Complexity. We now state the iteration and oracle complexity bound for TOBYQA.

THEOREM 4.10. (Oracle call complexity.) *Let Assumptions 4.1–4.8 hold. Let $\varepsilon > 0$ be any target gradient accuracy with $\varepsilon \leq 8L_g^2/\sigma_0$ exceeding the statistical noise resolution floor*

$$(4.15) \quad \varepsilon \geq \varepsilon_{\text{noise}} := C_0 \kappa_{eg} \max\left\{\delta_{\text{hi}}, \frac{\sigma_\varepsilon}{\delta_{\text{lo}}}\right\},$$

where $C_0 > 0$ is an absolute constant. Then the total number of iterations K_ε (and consequently the number of oracle evaluations after initialization) required to achieve $\|\nabla f(\mathbf{x}_k)\|_2 \leq \varepsilon$ satisfies

$$(4.16) \quad K_\varepsilon = O\left(\frac{L_g(f(\mathbf{x}_0) - f_{\text{low}})}{\varepsilon^2}\right).$$

The restriction $\varepsilon \geq \varepsilon_{\text{noise}}$ reflects the statistical resolution of single noisy queries: without replication, reductions below the observation-noise scale cannot be reliably distinguished from random fluctuations. The posterior-covariance stopping rule in Algorithm 2.1 is calibrated to this same scale.

Proof. Let $k \in \mathcal{S}_K$ be an accepted iteration prior to termination, which implies that $\|\nabla f(\mathbf{x}_k)\|_2 \geq \varepsilon$. Under Assumption 4.3 and condition (4.15), the gradient estimation error satisfies

$$\|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \leq \kappa_{eg} \zeta_k \leq \frac{\varepsilon}{C_0} \leq \frac{\|\nabla f(\mathbf{x}_k)\|_2}{C_0}.$$

Choosing $C_0 \geq 8$ gives $\|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \leq \frac{1}{8}\|\nabla f(\mathbf{x}_k)\|_2$, so by the reverse triangle inequality,

$$\|\mathbf{g}_k\|_2 \geq \|\nabla f(\mathbf{x}_k)\|_2 - \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \geq \frac{7}{8}\|\nabla f(\mathbf{x}_k)\|_2 \geq \frac{1}{2}\varepsilon.$$

Moreover, the model-error contribution satisfies

$$\begin{aligned} \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \|\mathbf{s}_k\|_2 &\leq \frac{\varepsilon}{C_0} \|\mathbf{s}_k\|_2 \leq \frac{2\|\mathbf{g}_k\|_2}{C_0} \|\mathbf{s}_k\|_2 \\ &= \frac{2}{C_0} \text{Pred}_k \leq \frac{1}{4} \text{Pred}_k. \end{aligned}$$

Since $\sigma_k \geq \sigma^*(\mathbf{g}_k)$ by Assumption 4.8, Lemma 4.5 guarantees that

$$f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{s}_k) \geq \frac{3}{4} \text{Pred}_k - \|\mathbf{g}_k - \nabla f(\mathbf{x}_k)\|_2 \|\mathbf{s}_k\|_2 \geq \frac{1}{2} \text{Pred}_k.$$

On an accepted iteration $k \in \mathcal{S}_K$, the iterate updates to $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{s}_k$. Thus, using $\text{Pred}_k = \|\mathbf{g}_k\|_2^{3/2} \sigma_k^{-1/2} \geq (\varepsilon/2)^{3/2} \sigma_{\max}^{-1/2}$ and the upper bound $\sigma_{\max} = \gamma_2 \frac{8L_g^2}{\varepsilon}$ from (4.13), the objective reduction satisfies

$$(4.17) \quad f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \geq \frac{1}{4} \left(\frac{\varepsilon}{2}\right)^{3/2} \sqrt{\frac{\varepsilon}{8\gamma_2 L_g^2}} \geq c \frac{\varepsilon^2}{L_g},$$

where $c = (32\sqrt{\gamma_2})^{-1} > 0$ is an absolute constant.

On rejected iterations $k \in \mathcal{U}_K$, the iterate remains unchanged ($\mathbf{x}_{k+1} = \mathbf{x}_k$), so $f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) = 0$. Summing the descent inequality (4.17) over all accepted iterations up to iteration K yields

$$f(\mathbf{x}_0) - f_{\text{low}} \geq f(\mathbf{x}_0) - f(\mathbf{x}_K) = \sum_{k \in \mathcal{S}_K} (f(\mathbf{x}_k) - f(\mathbf{x}_{k+1})) \geq |\mathcal{S}_K| c \frac{\varepsilon^2}{L_g},$$

whence the total number of accepted iterations satisfies

$$(4.18) \quad |\mathcal{S}_K| \leq \frac{1}{c} \frac{L_g (f(\mathbf{x}_0) - f_{\text{low}})}{\varepsilon^2} = O\left(\frac{L_g (f(\mathbf{x}_0) - f_{\text{low}})}{\varepsilon^2}\right).$$

By Proposition 4.9, the number of unsuccessful iterations is bounded by

$$|\mathcal{U}_K| \leq \frac{\log \gamma_1}{\log \gamma_2} |\mathcal{S}_K| + \frac{1}{\log \gamma_2} \log \frac{\sigma_{\max}}{\sigma_0}.$$

Since $\sigma_{\max}/\sigma_0 = \gamma_2(8L_g^2)/(\sigma_0\varepsilon) = O(\varepsilon^{-1})$, the additive logarithmic term satisfies

$$\frac{1}{\log \gamma_2} \log \frac{\sigma_{\max}}{\sigma_0} = O(\log(\varepsilon^{-1})),$$

which is asymptotically dominated by $O(\varepsilon^{-2})$. Combining the bounds on successful and unsuccessful iterations gives

$$K_\varepsilon = |\mathcal{S}_K| + |\mathcal{U}_K| \leq \left(1 + \frac{\log \gamma_1}{\log \gamma_2}\right) |\mathcal{S}_K| + O(\log(\varepsilon^{-1})) = O\left(\frac{L_g (f(\mathbf{x}_0) - f_{\text{low}})}{\varepsilon^2}\right).$$

Since TOBYQA evaluates one new query per iteration after the initial $2n+1$ evaluations, the total number of oracle calls satisfies $N_{\text{eval}} = 2n+1 + K_\varepsilon = O(L_g(f(\mathbf{x}_0) - f_{\text{low}})\varepsilon^{-2})$, completing the proof. $\square$

Remark 4.11. (Comparison with first-order complexity.) The rate $O(\varepsilon^{-2})$ in Theorem 4.10 matches the information-theoretic lower bound for first-order methods on $C^{1,1}$ nonconvex objectives [3]. This is consistent with the information available to TOBYQA at the budget $m = 2n+1$: the active set supports a fully-linear model, but not an asymptotically accurate Hessian. The $O(\varepsilon^{-3/2})$ rate of derivative-based adaptive cubic regularization [17, 5, 6] requires $\|(H_k - \nabla^2 f(\mathbf{x}_k))\mathbf{s}_k\|_2 = O(\|\mathbf{s}_k\|_2^2)$ [5, AM.4], whose recovery would generally require $O(n^2)$ samples. Under temporal drift, such a sample set also spans $O(n^2)$ time steps and incurs the nonlinear drift error quantified in Theorem 3.7. TOBYQA instead uses an $O(n)$ active set, with a shorter temporal span and the first-order complexity rate.

TABLE 5.1

Observation drift channels used in the benchmark experiments. Here $s_0 = \max(|f(\mathbf{x}_0)|, 1)$, t is the sequential oracle query index, and the Lorenz trajectory is generated with standard parameters $(s, r, b) = (10, 28, 8/3)$, initial state $(1, 1, 1)$, and integration step 0.01.

Channel	Analytical Form $\phi(f(\mathbf{x}), t)$	Classification
Static	$f(\mathbf{x})$	Baseline ($\beta = 0$)
LinearDrift	$f(\mathbf{x}) + 0.01 s_0 t$	Affine drift (Theorem 3.5)
PeriodicAdd	$f(\mathbf{x}) + 0.05 s_0 \sin(2\pi t/50)$	C^2 bounded drift (Theorem 3.7)
ChaoticDrift	$f(\mathbf{x}) + 0.03 s_0 \text{Lorenz}_x(t)$	Chaotic dynamic drift
Hetero	$f(\mathbf{x}) + 0.01 s_0 t (1 + \tanh \ \mathbf{x} - \mathbf{x}_0\)$	Spatially coupled drift
MultCoupling	$f(\mathbf{x}) \cdot (1 + 10^{-4}t)$	Multiplicative drift
PeriodicMult	$f(\mathbf{x}) \cdot (1 + 0.05 \sin(2\pi t/50))$	Multiplicative oscillatory drift

TABLE 5.2

Solve rates (%). Budget: 1500 evaluations per instance.

Solver	$\tau = 10^{-1}$			$\tau = 10^{-3}$			$\tau = 10^{-5}$			$\tau = 10^{-7}$		
	$n=6$	$n=10$	$n=20$	$n=6$	$n=10$	$n=20$	$n=6$	$n=10$	$n=20$	$n=6$	$n=10$	$n=20$
TOBYQA	91.4	88.6	87.3	70.2	59.9	41.7	34.4	32.0	26.2	28.3	27.7	24.0
NEWUOA	71.6	60.8	55.9	20.5	17.5	13.8	8.3	10.4	9.5	4.9	5.8	7.8
BOBYQA	61.9	57.6	50.9	18.3	15.6	10.1	7.6	9.2	7.7	4.1	5.6	6.0
Py-BOBYQA	74.8	60.7	41.4	46.9	27.3	10.3	24.4	14.2	7.0	21.1	12.5	6.5
Nelder–Mead	34.2	19.8	11.7	8.1	3.6	3.0	3.2	2.2	2.8	1.4	1.0	1.2
BFGS (FD)	13.7	10.0	0.9	1.0	0.4	0.7	0.5	0.4	0.7	0.4	0.4	0.7

5. Numerical Experiments. Experimental Setup. The benchmark testbed comprises 65 scalable smooth unconstrained test problems from the Moré–Wild benchmark set [15], evaluated at dimensions $n \in \{6, 10, 20\}$ with 5 random seeds controlling the observation noise realizations. Each problem is queried through one of the seven evaluation channels summarized in Table 5.1. Channel amplitudes and observation noise variances are scaled proportionally to the initial function magnitude $s_0 = \max(|f(\mathbf{x}_0)|, 1)$, with relative noise scale $\sigma_\varepsilon = 10^{-3}s_0$. The temporal query index advances as $t \leftarrow t + 1$ on each oracle evaluation across all solvers.

We benchmark TOBYQA against representative classical and modern derivative-free solvers:

- (i) **NEWUOA & BOBYQA** [19, 20]: Powell’s classical quadratic interpolation solvers, executed via the PDFO library [21];
- (ii) **Py-BOBYQA** [4]: A modern noise-aware model-based derivative-free solver;
- (iii) **Nelder–Mead** [16]: The classical heuristic simplex search method;
- (iv) **BFGS (FD)**: Quasi-Newton BFGS using forward-difference gradient approximations ($h = 10^{-4}$).

Every solver is allocated a maximum budget of 1500 evaluations per instance. Performance is measured by evaluating every queried point on the noise-free static latent objective f . Following Moré and Wild [15], an instance is declared solved at accuracy tolerance $\tau \in \{10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}\}$ if $\min_k f(\mathbf{x}_k) \leq f_L + \tau(f(\mathbf{x}_0) - f_L)$, where f_L is the global minimum achieved across all solvers.

5.1. Benchmark Comparison with Existing Solvers. Table 5.2 reports the overall solve rates across dimensions $n \in \{6, 10, 20\}$ and target tolerances $\tau \in \{10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}\}$. Figure 5.1 displays the cumulative data profiles at toler-

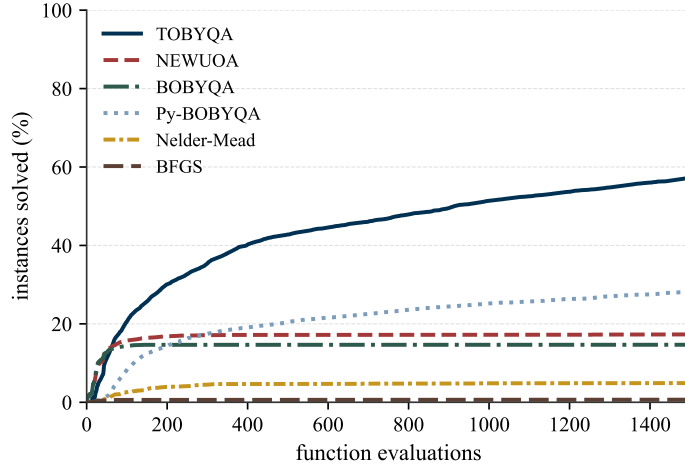


FIG. 5.1. Data profiles at tolerance $\tau = 10^{-3}$: fraction of benchmark instances solved as a function of the number of function evaluations, pooled across all test problems, dimensions, and drift channels.

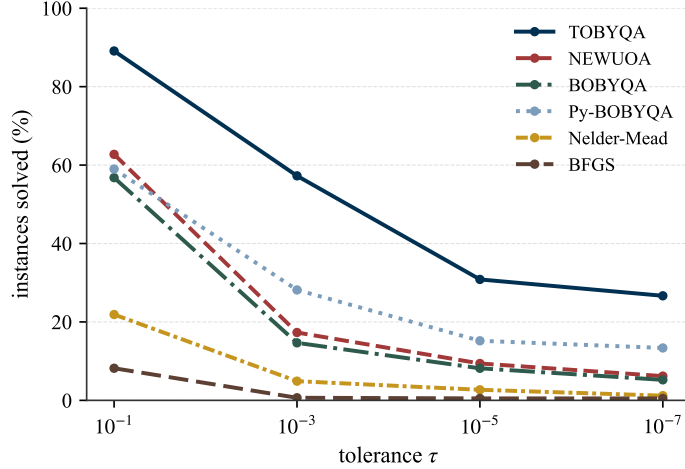


FIG. 5.2. Solve rates as a function of convergence tolerance $\tau \in [10^{-7}, 10^{-1}]$, pooled across all benchmark configurations.

ance $\tau = 10^{-3}$, and Figure 5.2 depicts the solve rate decay curves across tolerances spanning seven orders of magnitude.

The numerical results show the following patterns:

- (i) **Solve rates across tolerances:** TOBYQA has the highest solve rate at every reported tolerance and dimension (Table 5.2). At $\tau = 10^{-3}$, TOBYQA solves 70.2% of instances at $n = 6$, 59.9% at $n = 10$, and 41.7% at $n = 20$, compared with 46.9%/27.3%/10.3% for Py-BOBYQA, 20.5%/17.5%/13.8% for NEWUOA, and 18.3%/15.6%/10.1% for BOBYQA.
- (ii) **Stationary interpolation versus temporal drift:** In the stationary environment (Static in Table 5.3), Py-BOBYQA attains a 59.3% solve rate, closely

TABLE 5.3

Solve rates (%) at $\tau = 10^{-3}$ broken down by drift regime, pooled over all n and seeds.

Solver	Chaotic	Hetero	Linear	MultCoupling	PeriodicAdd	PeriodicMult	Static
TOBYQA	56.0	50.1	64.9	56.5	56.2	59.2	57.9
NEWUOA	10.4	11.6	10.1	29.4	14.7	20.1	24.9
BOBYQA	12.4	9.1	9.6	21.0	12.3	17.3	20.8
Py-BOBYQA	14.2	3.0	4.0	55.4	19.3	42.1	59.3
Nelder–Mead	1.1	0.0	0.6	12.3	3.1	6.3	10.9
BFGS (FD)	1.9	0.0	0.4	0.3	0.8	0.8	0.3

TABLE 5.4

Budget utilisation. A solver that stops before the budget is exhausted has declared convergence; under noise this is the dominant failure mode of heuristic stopping rules.

Solver	median evals used	% of budget	% runs stopping early
TOBYQA	1500	100.0	8.5
NEWUOA	185	12.3	99.0
BOBYQA	147	9.8	99.8
Py-BOBYQA	1500	100.0	24.2
Nelder–Mead	1500	100.0	0.0
BFGS (FD)	180	12.0	99.6

matching TOBYQA (57.9%), while NEWUOA and BOBYQA reach 24.9% and 20.8%. Under temporal drift, however, stationary model-based methods deteriorate sharply: Py-BOBYQA drops to 4.0% on LinearDrift and 3.0% on Hetero (versus 64.9% and 50.1% for TOBYQA), while NEWUOA and BOBYQA achieve only 9%–15% across additive drift channels. In these methods, unmodeled temporal drift enters the spatial quadratic model and distorts the search direction.

- (iii) **Finite-difference BFGS under drift:** BFGS (FD) has low solve rates under the drifting, noisy channels. With $\sigma_\varepsilon = 10^{-3}s_0$ and finite-difference step $h = 10^{-4}$, the difference quotients amplify observation noise and incur an $O(\beta/h)$ bias under affine drift.
- (iv) **Evaluation-budget use:** As shown in Table 5.4, NEWUOA and BOBYQA return before exhausting the budget on 99.0% and 99.8% of runs, with median totals of 185 and 147 evaluations out of 1500. Py-BOBYQA stops early on 24.2% of runs with a median of 1500 evaluations. In contrast, TOBYQA terminates early on only 8.5% of runs, utilizing the available budget when progress continues and stopping via the posterior-covariance criterion (2.19) when the estimated gradient uncertainty matches the noise threshold.

5.2. Ablation Studies and Curvature Treatment. Table 5.5 reports an ablation study across the seven evaluation channels.

Impact of Drift Compensation. In this ablation, the time-augmented KKT system is retained for spatial-gradient recovery, but its recovered coefficient is set to $\gamma = 0$ in the acceptance statistic (2.16). The ablated variant is nearly unchanged on the Static channel (56.0% vs 57.9%), whereas its solve rate falls to 7.1%–12.7% across the additive drifting channels. The comparison isolates the practical contribution of drift compensation in the acceptance test.

Role of the Curvature Prior Kernel A. Replacing the composite kernel $K =$

TABLE 5.5

Ablation of TOBYQA components: solve rates (%) at $\tau = 10^{-3}$ by drift regime, pooled over all n and seeds.

Solver	Chaotic	Hetero	Linear	MultCoupling	PeriodicAdd	PeriodicMult	Static
TOBYQA	56.0	50.1	64.9	56.5	56.2	59.2	57.9
+ interpolated H	47.4	55.1	66.4	61.2	50.6	63.9	59.5
– time column	9.6	7.1	8.5	55.5	12.7	46.2	56.0
– curvature kernel A	29.0	30.1	39.9	35.5	34.6	36.3	34.5
+ Powell update	11.9	18.7	16.9	14.4	13.4	14.8	14.5
frozen σ	39.1	36.3	38.4	37.6	38.3	36.1	36.2

TABLE 5.6

Solve rates (%) of the Zero-Hessian default versus the interpolated Hessian, by dimension and tolerance. The first three column groups use the main budget of 1500 evaluations; the last two use budget-scaled cells ($n = 6$ with 450 evaluations, $n = 20$ with 5000). Pooled over all 65 problems, 7 drift channels, and 5 seeds.

Solver	$n = 6$		$n = 10$		$n = 20$		$n = 6, b=450$		$n = 20, b=5000$	
	10^{-3}	10^{-5}	10^{-3}	10^{-5}	10^{-3}	10^{-5}	10^{-3}	10^{-5}	10^{-3}	10^{-5}
TOBYQA ($H_k \equiv 0$)	70.2	34.4	59.9	32.0	41.7	26.2	83.3	64.3	67.1	45.9
+ interpolated H	62.4	24.0	60.1	29.9	50.7	38.9	67.3	44.9	70.8	59.7

$A + \rho^{-1}I_m$ with an unweighted ridge penalty $K = \rho^{-1}I_m$ (which treats residuals as independent white noise and ignores curvature correlation) reduces the solve rate from 57.9% to 34.5% on Static and from 64.9% to 39.9% on LinearDrift. This result is consistent with the statistical interpretation in Proposition 3.9: the kernel $A_{ij} = \frac{1}{2}(\mathbf{d}_i^\top \mathbf{d}_j)^2$ accounts for correlated residual variation induced by unmodeled curvature. **Curvature Treatment: Zero or Interpolated Hessian.** The Zero-Hessian default of Section 2.2 was adopted on identifiability grounds: at $m = 2n+1$ the $\frac{1}{2}n(n+1)$ quadratic coefficients are underdetermined. The benchmark data show, however, that the relative merit of the interpolated Hessian $H_k = \sum_i \lambda_i \mathbf{d}_i \mathbf{d}_i^\top$ varies with dimension in a way that this argument alone does not predict. Table 5.6 isolates the two variants. At $n = 6$ the default is better at both tolerances (70.2% vs 62.4% at $\tau = 10^{-3}$; 34.4% vs 24.0% at $\tau = 10^{-5}$), and the gap widens at the reduced budget of 450 evaluations (83.3% vs 67.3%). At $n = 20$ the ordering reverses at every tolerance (50.7% vs 41.7% at $\tau = 10^{-3}$; 38.9% vs 26.2% at $\tau = 10^{-5}$), and at the enlarged budget of 5000 evaluations the interpolated variant also wins the paired comparison (1229 of 2275 instances against 963 for the default, 83 ties). Because the two budget-shifted cells align the per-variable evaluation budget across dimensions, the reversal follows the dimension rather than the budget.

The environment breakdown at $n = 20$ and budget 5000 shows a drift-specific pattern: at $\tau = 10^{-3}$, the interpolated Hessian achieves higher solve rates on Static, Hetero, and LinearDrift (84.4% vs 60.8% on Static), but lower rates on ChaoticDrift and PeriodicAdd (54.5% vs 72.1% and 49.8% vs 75.7%, respectively; at $\tau = 10^{-5}$, the corresponding rates are 39.0% vs 67.2% and 38.2% vs 68.0%). The latter two channels contain nonlinear additive drift, which contributes to the recovery error bounded in Theorem 3.7. The identifiability ratio $(2n+1)/(\frac{1}{2}n(n+1))$ decreases monotonically in n , yet the interpolated Hessian becomes more useful as n grows, so underdetermination alone does not explain the reversal. We therefore retain the Zero-Hessian default and leave the dimension-dependent role of interpolated curvature as an open question

(Section 6). Powell’s incremental update has solve rates of 12% $\sim$ 19% across all channels (Table 5.5), consistent with the effect of carrying curvature information across changing temporal baselines.

5.3. Numerical Checks of the Drift-Recovery Results. We conclude the experiments with two numerical checks of the results in Section 3.

Affine-drift gradient invariance (Theorem 3.5). We generated synthetic sample sets $\mathcal{Y}$ and added linear drift at rates $\beta \in \{1, 10^3, 10^6\}$. Relative to the drift-free baseline, the spatial gradient difference $\|\mathbf{g} - \mathbf{g}^{\text{stat}}\|_2$ was 1.5×10^{-13} at $\beta = 1$ and 5.5×10^{-8} at $\beta = 10^6$, while the recovered temporal coefficient agreed with $\gamma^{\text{stat}} + \beta$ to machine precision across all tested configurations.

Nonlinear-drift bias (Theorem 3.7). Under quadratic drift $\mu(t) = \frac{1}{2}L_\mu t^2$, we evaluated $|\gamma - \gamma^{\text{stat}} - \mu'(t_k)|$ as a function of the active-set temporal span τ_{max} . Over an eightfold variation in τ_{max} , the empirical bias was approximately linear in τ_{max} , with correlation $r > 0.998$, consistent with the bound for the recovered temporal coefficient in Theorem 3.7.

6. Conclusion. We have introduced TOBYQA, a time-augmented framework for derivative-free optimization under noise and temporal drift.

The analysis establishes solvability of the augmented KKT system under a full-column-rank condition on X_{aug} . Because the composite kernel $K = A + \rho^{-1}I_m$ is positive definite, this condition is sufficient for a unique solution without the strict interpolation poisedness required by classical minimum-Frobenius-norm models (Proposition 3.2). The same recovery operator satisfies $\Pi X_{\text{aug}} = I_{n+2}$. Consequently, under affine observation drift, the recovered spatial gradient is invariant to the drift magnitude, while under C^2 nonlinear drift its bias is bounded by $O(L_\mu \tau_{\text{max}}^2)$ and therefore depends on the temporal span of the active sample set (Theorems 3.5 and 3.7).

The kernel also admits a statistical interpretation. Under a rotation-invariant Gaussian curvature prior, Powell’s residual kernel $A_{ij} = \frac{1}{2}(\mathbf{d}_i^\top \mathbf{d}_j)^2$ is proportional to the covariance of the omitted quadratic forms, so the soft-constrained KKT estimate coincides with the generalized least-squares estimator under the resulting residual model (Proposition 3.9). Under the fully-linear framework and the regularization confinement condition, the oracle complexity above the statistical noise floor is $O(L_g(f(\mathbf{x}_0) - f_{\text{low}})\varepsilon^{-2})$, matching the lower bound for first-order nonconvex optimization (Theorem 4.10).

The experiments cover 65 benchmark problems across three dimensions ($n \in \{6, 10, 20\}$) and seven evaluation channels. TOBYQA has the highest reported solve rates in these tests, with the largest differences occurring under additive drift. The ablations show lower solve rates when either drift compensation or the curvature prior kernel is removed. The evaluation-budget comparison also shows that the effect of fitting explicit quadratic curvature depends on both dimension and drift channel: it is less effective in low dimension and under nonlinear drift, but can be beneficial in higher dimension under the tested static and affine-drift settings. This pattern is not explained by the identifiability ratio of the quadratic coefficients alone.

Several questions remain open. The dimension-dependent behavior of the interpolated curvature variant calls for a criterion that predicts when curvature estimation is useful at the budget $m = 2n + 1$. The present framework assumes a static latent objective with a drifting observation channel; extending it to $f(\mathbf{x}, t)$ would require a model for the motion of $\mathbf{x}_*(t)$ and corresponding tracking-error bounds. Polynomial time columns $[\boldsymbol{\theta}, \boldsymbol{\theta}^2, \dots, \boldsymbol{\theta}^p]$ could be used to represent polynomial drift, but the conditioning of the enlarged system and its effect on recovery error remain to be

analyzed. Extensions to bound constraints $\mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$ and general linear constraints provide another direction for further work.

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